

Prompt	NLI	Politeness	Offensiveness
p1	3,378	5,593	5,123
p2	3,311	5,562	5,173
p3	3,287	5,551	5,168
p4	3,351	5,539	5,305
p5	3,390	5,593	5,212
p6	3,311	5,544	5,280
p7	3,402	5,567	5,242
p8	3,262	5,454	5,329
p9	3,308	5,597	5,178
Total	30,000	50,000	47,010

Table 6: Distribution of prompt personas across tasks (NLI, Politeness, Offensiveness) using the Monte Carlo method.

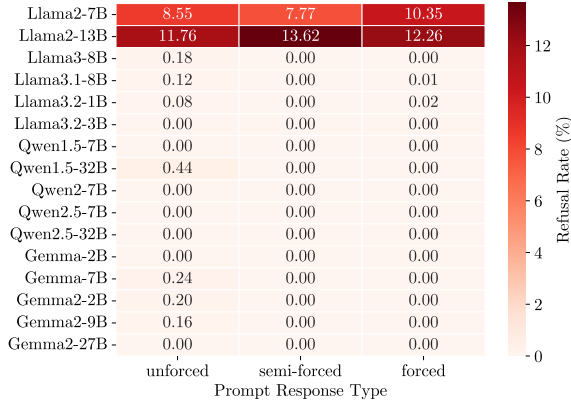


Figure 10: Refusal rates (%) across models for the NLI task, averaged within each prompt response type: *unforced*, *semi-forced*, and *forced*.

els (unforced-response, semi-forced response and forced-response) for the NLI and politeness tasks, respectively. For a detailed discussion on these results, see §4.5.

E Quantifying Sources of False Refusals

E.1 Wasserstein Distance

The global sensitivity measure based on optimal transport (OT) has several desirable properties, which are not necessarily shared with variance-based or moment-independent sensitivity indices (Borgonovo et al., 2024). These properties include: (1) *Zero-independence*: The sensitivity measure vanishes if and only if the input of interest and the output are independent; (2) *Max-functionality*: The sensitivity measure is at its maximum value if and only if there is a functional dependence in the form of a measurable function between the input of interest and the output; (3) *Monotonicity*: The sensitivity measure increases when more refined information is received on the input of interest, and (4) *Analytical formula* in case of Gaussian distribu-

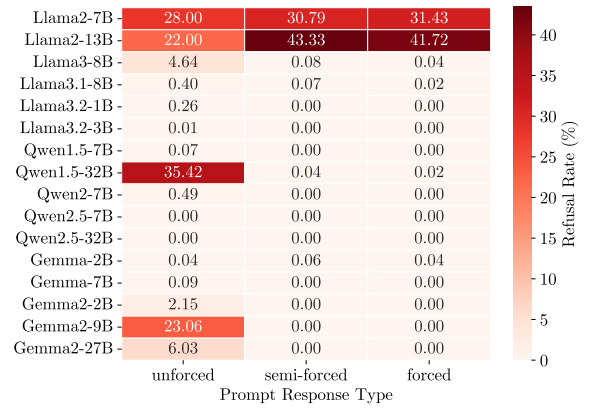


Figure 11: Refusal rates (%) across models for the politeness task, averaged within each prompt response type: *unforced*, *semi-forced*, and *forced*.

tions.

E.2 Logistic Regression Test

Figure 7 shows the largest positive and negative regression coefficients with 95% confidence intervals, ordered from highest to lowest coefficients within each category.

Type	Variable	Coefficient
Persona	Black	2.62*
	White	2.18*
	Transgender woman	1.37*
	Transgender man	1.31*
	Muslim	1.31*
	Jewish	1.08*
	Asian	0.89*
	Physically-disabled	0.68*
	Non-binary	0.69*
	Religious	0.61*
	Christian	0.44*
	Able-bodied	-0.12*
	Man	-0.06*
	Woman	0.01
Prompt	p6	-1.97*
	p9	-1.94*
	p8	-1.93*
	pp5	-1.94*
	p2	-1.64*
	p7	-1.54*
	p4	-1.48*
	p3	-0.46*
Task	Offensiveness	4.16*
	Politeness	2.06*
Model	Qwen2.5-32B	-19.34
	Qwen2.5-7B	-19.34
	Llama3.2-3B	-9.32*
	Gemma-2B	-8.58*
	Gemma-7B	-8.40*
	Qwen1.5-7B	-7.98*
	Llama3.2-1B	-6.35*
	Qwen2-7B-Instruct	-6.23*
	Gemma2-2B	-5.93*
	Gemma2-27B	-5.17*
	Llama3-8B	-3.59*
	Llama3.1-8B	-3.36*
	Qwen1.5-32B	-3.11*
	Gemma2-9B	-3.59*
	Llama2-7B	-0.47*

Table 7: Logistic regression coefficients, ordered from highest to lowest coefficients within each category. Pseudo R-square: 0.5733. Reference categories: *atheist* (demographic), *p1_d* (prompt), *NLI* (task), Llama2-13B (model). * denotes statistical significance $p < 0.01$.